[METRIC]
A-A-3059
January 29, 1997
Superseding
TT-P-54C
March 27, 1973

COMMERCIAL ITEM DESCRIPTION

PAINT: PHOTOLUMINESCENT, NONRADIOACTIVE, INTERIOR

The General services Administration has authorized the use of this commercial item description by all federal agencies.

1. **SCOPE.** This commercial item description describes three grades of interior, latex, phosphorescent paint. These paints are nonflammable, nonradioactive, and meet the luminance requirements of the International Maritime Organization (IMO) for Passenger Ships. They are capable of being applied to paint primers and topcoats and are visibly luminescent in the dark for a minimum of six hours.

2. **Classification.** The phosphorescent paints described here shall be of three grades, as follows:

Grade A - Yellow-Green, high initial brightness, long afterglow
Grade B - Yellow-Green, moderate afterglow
Grade C - Blue-Green, moderate afterglow

3. **SALIENT CHARACTERISTICS.**

3.1 **General requirements.** The paints shall be a one component, ready-to-use, latex paint. They shall be suitable for application by brush, roller, and spray, and capable of being top coated with a clear varnish.

3.1.1 **Prohibited materials.** The manufacturer shall ensure that no mercury, cadmium, hexavalent chromium compounds, compounds containing pyrophosphates, Hazardous Air Pollutants (HAPs), known or suspected human carcinogens, (as defined by the National Toxicology Program's Annual Report on Carcinogens), toxic pollutants, or Ozone Depleting Substances (ODS) are used in the formulation. If any of these substances is present as an impurity in a raw ingredient, its concentration shall be less than 0.1 percent by weight. The lead content of the nonvolatile portion of the coating shall not exceed 0.06 percent. Radioactive materials shall not be used in the formulation. The radioactivity of the paint, as measured by a Geiger counter, shall be the same as the background radioactivity.

3.1.2 **Condition in container.** The coating shall be free from skins, livering, seeds, hard settled pigment, and shall be readily dispersible to a uniform condition by five minutes of hand stirring. A closed, three-quarter filled container shall not skin within 48 hours, when stored at room temperature.

3.2 **Special requirements.** Unless otherwise stated in the test method, all routine and referee testing shall be done at the conditions specified in ASTM D 3924

3.2.1 **Phosphorescence.** Two panels prepared as describe in note 1, with the exception that they shall be disks whose diameter are 35-50 mm (1-3/8 - 2 inches), shall be placed in a completely dark room for 24 hours.
They shall then be irradiated with a cool white fluorescent light producing 900 lx, at a distance of 0.4 m (16

Beneficial comments, recommendations, additions, deletions, clarifications, etc. and any data which may improve this document should be sent to: General Services Administration, Paints and Chemicals Center, Engineering and Commodity Management Division (10FTE), 400 15th St. SW, Auburn, WA. 98001.

FSC 8010

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inches). The panels shall be irradiated for five minutes. The room shall then be made completely dark. Luminance intensity shall then be measured at a distance from the disks equal to the diameter of the disks. The light aperture of the photometer head shall be no greater than 10 mm (0.44 inches). The intensity values obtained shall meet the requirements of table I. In cases of dispute, a 150 watt, xenon light with a wavelength range of 380 to 450 nm, shall be used instead of the cool white fluorescent light.

3.2.2 **Quantitative requirements.** The phosphorescent paint shall meet the requirements listed in table II.

Table I. Minimum Luminance of Phosphorescent Coating

Time after excitation	Grade A	Grade B	Grade C
Luminosity in mcd/m^2, min.			
10 minutes after excitation	30	15	15
60 minutes after excitation	2.2	2.2	2.0
120 minutes after excitation	1.2	---	---
320 minutes after excitation	0.32	0.2	0.2
Color emission	Yellow-Green	Yellow-Green	Blue-Green

TABLE II

QUANTITATIVE REQUIREMENTS

PROPERTY	REQUIREMENT	ASTM TEST METHOD
Prohibited materials		
Lead content, wt. percent of nonvolatile, max	0.06	See note 4
Other prohibited materials listed in 3.1.2, wt. percent	0.1	See note 4
Flash point, degrees C. (Degrees F.), min	93 (200)	D 56
Drying time, hours		
Dry to touch, max	1	D 1640 1/
Dry to recoat, within	8	D 1640 1/
Fungus resistance, min.	8	D 3273, D 3274
Volatile Organic Compound (VOC) content (less water and exempt solvents), g/l (lb/gal), max	250	D 3960 2/
Freeze-thaw resistance, three cycles.		
Difference from the control specimen	None	D 2243
Maximum viscosity difference, K.U.	8	D 2243 and D 562
Adhesion	5B	D 3359 1/, 3/
Odor	Not obnoxious	D 1296
Package stability of paint		
Viscosity after storage, K.U., max.	100	D 1849 and D 562
Settling after storage	8	D 1849 and D 869

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1/ The panels used shall be gypsum wallboard conforming to ASTM C 36. The wallboard shall be coated with a white primer. The primer shall be allowed to dry for 24 hours at room temperature. Two coats of phosphorescent paint shall then be applied in accordance with the manufacture's instructions. After allowing the phosphorescent paint to dry in accordance with the manufacture's instructions, the coating shall be topcoated with a spar varnish. For referee purposes, the white primer and spar varnish shall meet the requirements of A-A-2994 and A-A-1788, respectively.

2/ The VOC shall be determined on the paint as applied in accordance with the manufacturer's instructions for use.

3/ Use test method B. Allow the coating to dry for 72 hours at room temperature prior to testing. Use Scotch Brand Tape number 810 or any other cellophane tape with the same adhesive strength.

4/ For referee purposes only. Lead content shall be measured using X-Ray fluorescence, cadmium content by ASTM D 3335, chromium content by ASTM D 3718, and mercury content by ASTM D 3624. Organic solvents shall be identified using FED-STD-141 methods 7356 and 7375.

4. REGULATORY REQUIREMENTS.

4.1 **Federal Acquisition Regulations (FAR).** The offeror/contractor is encouraged to use recovered materials to the maximum extent practicable, in accordance with paragraph 23.403 of the FAR.

4.2 **Code of Federal Regulations (CFR).** The paint shall not contain any substance listed in the following Code of Federal Regulations as a hazardous air pollutant, toxic pollutants, or ozone depleting substance:

a. 40 CFR part 61.
b. 40 CFR part 401
c. 40 CFR part 82

4.3 **MSDS.** The manufacturer shall comply with requirements set forth by the Hazardous Communication Standard 29 CFR 1910.1200 (d) through (g). All Material Safety Data Sheets (MSDSs) submitted shall comply with provisions of FED-STD-313.

5. QUALITY ASSURANCE PROVISIONS.

5.1 **Product conformance.** The contractor shall maintain substantiating evidence that the product offered meets the salient characteristics of this Commercial Item Description and that the product conforms to the producer's own drawings, specifications, standards, and quality assurance practices, and is the same product offered for sale in the commercial marketplace.

The contractor shall provide the required information in a tabulated format and with enough clarity so that the formulation of the tested product can be traced compared to the offered product(s). The contractor shall also provide a summary of performance data, consisting of test reports, substantiating that the product to be supplied under this CID meets the requirements cited in table II and is the same product offered for sale in the commercial marketplace. Referee test data does not need to be provided.

The government reserves the right to require proof of such conformance prior to first delivery and thereafter as may be otherwise provided for under the provisions of the contract.

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5.2 **Luminance.** The contractor shall also provide test data from an independent, government approved laboratory which shows that the paint meets the luminance requirements in 3.2.1.

6. **PACKAGING.**

Preservation, Packing, and marking shall be as specified in the contract or order.

7. **NOTES.**

7.1 **Intended use.** The phosphorescent paint described in this commercial item description is a high grade material designed for use as markers, indicators, etc., in civilian and non-operational military applications. All grades meet luminescent requirements for photoluminescent systems of the International Maritime Organization, adopted in 1993. Operational military uses are covered by A-A-50598.

7.1.1 **Application.** This paint is intended to be applied over properly primed steel, wood, and wallboard, and topcoated with a clear, washable, top coat. The primer should be white.

7.1.2 **Other intended applications.** This phosphorescent paint also can be used in interior accommodations, working and machinery spaces of merchant vessels. The markings applied should be sufficient in character to clearly show in darkness the positions of exit doors, ports, ladders, companionways, the location of emergency lights, control valves, and similar vital locations. If a fire retardant phosphorescent paint is required, use paint conforming to A-A-50598.

7.1.3 **Grades.** Grade A paint has a high initial brightness and exceeds the requirements of the German Industrial Norm (DIN) 67510. It's luminance intensity after five hours is approximately 100 times the visibility limit. Grade B meets the requirements of DIN 67510. Grade C has a little lower luminance intensity than Grade B.

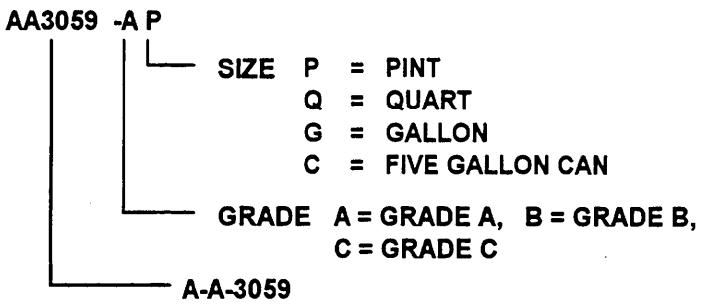
7.2 **Ordering Data.** Purchasers should include the following information in the contract or purchase order:

(a) Title, number, and date of this commercial item description
(b) Grade required
(c) Quantity and size of the container required
(d) Address to whom MSDSs should be sent
(e) Packaging, packing, and marking required.

7.2.1 **Evaluation of bids.** Since the luminescence is to be determined at the coverage recommended by the paint manufacture, procurement officers should evaluate bids by the cost of paint per square foot of coverage.

7.3 **Part Identification Number (PIN).** The following part identification numbering procedure is for government purposes and does not constitute a requirement for the contractor.

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7.4 **Luminance units.** The currently used units of luminance in industry is the millicandela per square meter (mcd/m^2). The conversion from microlamberts, use in TT-P-54C, to millicandela per square meter is:

mcd/m^2 = $(10/\pi)$ microlamberts.

7.5 **Relationship between this commercial item description and TT-P-54C.** The Grade A phosphorescent paint should be used for TT-P-54C type I material. Grade C should be used for TT-P-54C type II material. Grade C material has about a 20 percent decrease in luminance after four hours than TT-P-54C type II.

7.6 **Referenced documents.**

Federal Specifications:

A-A-1788 - Varnish, Oil: Interior.
A-A-2994 - Primer Coating, Interior, For Walls and Wood.
A-A-50598 - Kit, Photoluminescent Marking.

Federal Standards:

FED-STD-313 - Material Safety Data, Transportation Data and Disposal Data for Hazardous Materials furnished to Government Activities.

ASTM Standards:

C 36 - Gypsum Wallboard.
D 56 - Flash point by Tag Closed Tester.
D 562 - Consistency of Paints Using the Stormer Viscometer.
D 869 - Evaluating Degree of Settling of Paint.
D 1640 - Drying, Curing, or Film Formation of Organic Coatings at Room Temperature.
D 1849 - Package Stability of Paint.
D 2243 - Freeze-Thaw Resistance of Water-Borne Coatings.
D 3273 - Resistance to Growth of Mold on the Surface of Interior Coatings in an Environmental Chamber.
D 3274 - Evaluating Degree of Surface Disfigurement of Paint Films by Microbial (Fungal or Algal) Growth or Soil and Dirt Accumulation.
D 3359 - Measuring Adhesion by Tape Test.
D 1296 - Odor of Volatile Solvents and Diluents.

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D 3924 - Standard Environment for Condition and Testing Paint, Varnish, Lacquer, and Related Materials.
D 3960 - Determining Volatile Organic Compound (VOC) Content of Paints and Related Coatings.

National Toxicology Program

Annual Report on Carcinogens

7.7 **Source of Documents.**

7.7.1 Contact the contracting officer for a copy of paragraph 23.403 of the FAR,

7.7.2 Copies of ASTM specifications and standards may be obtained from the American Society for Testing and Materials, 100 Barr Harbor Drive, West Conshohocken, PA 19428-2959.

7.7.3 Contact the contracting officer for copies of the appropriate paragraphs in 29 and 40 CFR.

7.7.4 Copies of Federal Specifications and standards may be obtained from the Federal Supply Service Bureau, Specification Section, Suite 8100, 470 East L'Enfant Plaza, SW, Washington, DC 20407.

7.7.5 Copies of the Annual Report on Carcinogens may be obtained from the National Toxicology Program, Annual Report on Carcinogens, PO Box 12233, Research Triangle Park, NC 27709.

MILITARY INTERESTS:

Custodian
Air Force - 99
Coast Guard - CG

Review Activities
Army - MI, MR
Air Force - 84

User Activities
Navy - OS, AS, SA
Marines - MC

CIVIL AGENCY COORDINATING ACTIVITIES:

GSA - FSS
GSA - PQSB
HUD - HGP
VA - OSS
COM - NIST
INT - BOR
DOT - ACO
DOT - APM
DC - DCG

Preparing Activity:
GSA - FSS

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